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International Islamic University Chittagong
Department of Computer Science & Engineering
B. Sc. in CSE Semester Final Examination, Autumn-2024
Course Code: CSE2423 **Course Title: Database Management System**
Total marks: *50* Time: *2* hours *30* minutes

[Answer *all* the questions. Figures in the right hand margin indicate full marks.
Separate answer script must be used for Group A and Group B]

		CLO	BT	Marks

Group-A

1. a) Define encryption and decryption. Determine why roles are needed in case of privileges. Explain with appropriate assumption. — CLO1 A 4

b) Consider the following schemas for relations and answer the following questions — CLO2 A 6
Customer (customerID, cname, email, phone, address, orderID)
Order (orderID, productID, item, quantity, price, orderDate)
Product (productID, name, model)

i. Write the sql statement to create these relations, you have to include all integrity constraints.

ii. Define integrity constraints on Order relation where quantity must be less than 10.

iii. Determine the trigger statement for each order after the order placement of a product more than 5 quantity where you have to give 5% discount on that product. Keep the log track for this trigger to see the discount price of that order.

OR

b) Normalize the following table if it is not normalized:

pid	pname	emp_id	emp_name	dept_id	dept_name
P01	Alpha	E01	John	D01	HR
P02	Beta	E02	Alice	D02	Finance
P01	Alpha	E03	Mike	D01	HR

2. a) Consider the relation schema R(ABCDEFG) with the following functional dependencies: — CLO2 A 6

A -> C
B -> A
CD -> B
E -> F
G -> E

i. Determine if "CDG" and "BCE" are candidate keys. Justify your answer.
ii. Decompose the schema into BCNF and explain the decomposition process.

b) Explain the difference between dependency preservation and lossless decomposition with examples. — CLO2 A 4

OR

b) Consider a relation R(A, B, C, D, E, F)

F1={A=>BC, B=>CDE, AE=>F}
F2={A=>BCF, B=>DE, E=>AB}

Check whether F1 and F2 are equivalent or not.

Group-B

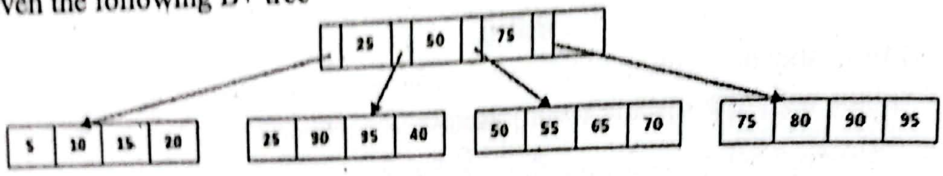
3. a) Compare and contrast dense and sparse indexing techniques. Provide scenarios where each would be more effective. CLO2 A

OR

a) Discuss the role of clustered indexes in improving query performance. Provide examples.

CLO2 A

b) Given the following B+ tree



Requirements:

1. Insert step-by-step: 22, 30, 35, 51, 72
2. Delete: 30, 50, 70

4. a) Consider the following three transactions and schedule (time goes from top to bottom). Is this schedule conflict-serializable? Explain why or why not. CLO2 A

Transaction T_0	Transaction T_1	Transaction T_2
$r_0[A]$		
$w_0[A]$		
		$r_2[A]$
		$w_2[A]$
	$r_1[A]$	
$r_0[B]$		
		$r_2[B]$
$w_0[B]$		
		$w_2[B]$
	$r_1[B]$	
	c_1	
c_0		
		c_2

b) Do you think conflict-serializability and view-serializability will be enough to make schedule recoverable and cascade-less? Justify your answer through an example. CLO2 A 5

OR

b) Explain how a transaction will be accomplished through different states. Also, illustrate which properties are mandatory in terms of database transactions.

5. a) Discuss the importance of distributed databases in modern systems. Explain the trade-offs between replication and fragmentation. CLO2 2 4

b) What is the role of logging in database recovery? CLO2 2 3

c) Discuss two techniques for resolving system failures in a database system. CLO2 2 3

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[The figures in the right hand margin indicate full marks. Course Outcomes and Bloom's Taxonomy Levels are mentioned in additional columns. **The questions must be answered in order.**]

Group A

		Marks	CO	DL
1) a)	Consider the following relational database: ***Patient*** (<u>pid</u>, pname, Address, mobile, DOB, gender) ***visit*** (pid, did, visit_date) ***Doctor*** (<u>did</u>, dname, speciality) **drug** (<u>drid</u>, d_name, d_type, manuf_year, unit_price) **prescribe** (pid, did, drid, pdate, quantity) Give an SQL DDL definition of this database. Identify referential-integrity constraints that should hold, and include them in the DDL definition. **Ensure the following constraints:** Pname, dname, and d_name (not null), Mobile (unique), gender ('M', or 'F'), quantity and unit_price (not negative). **OR** Normalize the following table if it is not normalized.	6	2	3

sid	sname	address	Course_id	cname	Department_id	dname	D_location
S01	Forhad	ctg	C01, C02	C, C++	D01	CSE	GEC
S02	Anika	dhaka	C01, C03	C, JAVA	D01	CSE	GEC
S03	Mazed	ctg	B01	Statistics	D02	BBA	Agrabad

		Marks	CO	DL
b)	How the trigger does ensure integrity of the database? Give example. **OR** Write an assertion for the patient database to ensure that the patient has not visiting the two doctors at a time. (May consider the schema of Q1 (a)).	4	2	2
c)	Let the relation schema, **R(ABCDEFGH)** have the following F set of functional dependencies: A->B CH->A B->E BD->C EG->H DE->F i. Can "CHDG" and "DEHC" be candidate keys for the above functional dependencies? Compute the closure and justify your answer. ii. The above schema is not in BCNF but in 2NF. Decompose the relation into BCNF and explain your answer.	2 + 3	2	4
d)	Why data normalization is important? Explain lossless decomposition and lossy decomposition with an example.	5	1	2

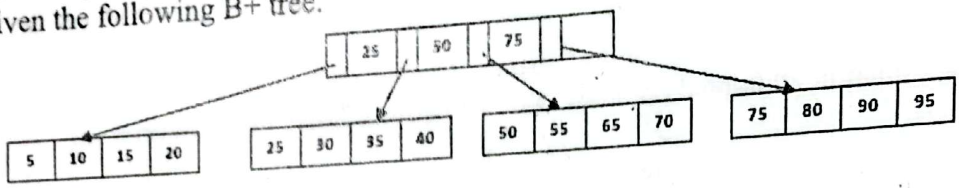
Group B

3.a) How to choose indexing or hashing technique? Justify. 4

OR

How can Indexes help performance? Consider employees relation, if we want to retrieve all employees, whose salary is in a given range, will it be best alternative to sort the employee records by employee id. Justify your answer. 6

3.b) Given the following B+ tree.



Requirements:

1. Insert (step by step): 13,12,17, 60, 45
2. Delete: 35, 60, 75, 95

4. i. For the following transaction T_i, transfers \$50 from account A to account B is shown below: 3+2+2+3

T_i: read(A);
A:=A-50;
write(A);
read(B);
B:=B+50;
write(B);

Considering the above example, explain each of the ACID properties.

ii. Explain the distinction between the terms "**serial schedule**" and "**serializable schedule**".

iii. *"A given schedule can be tested for conflict serializability by constructing a precedence graph for the schedule and by searching for the absence of cycles in the graph."* – State whether the statement is true or not using an example.

iv. What is a cascadeless schedule? Why is cascadelessness of schedules desirable? Are there any circumstances under which it would be desirable to allow non-cascadeless schedules? Explain your answer.

5.a) Why **distributed database** is important? *"Replication and fragmentation are the two ways of storing data in the distributed database"*. Distinguish between these two systems explaining the advantage of one over the other. 3

5.b) Explain the purpose of **checkpoint** mechanism. How often should checkpoints be performed give an example. 3

OR

Draw the **state diagram of a transaction** and elaborately discuss its state.

5.c) How can you resolve problem of system failure? Explain at least two techniques 4

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Time: 2 hours 30 minutes *Total marks: 50*

[Answer the following questions. Figures in the right hand margin indicate full marks.] CLO DL

Group A

[Answer the following questions]

a) Describe encryption decryption and privilege. 2 CLO1 C2

b) Given the following schema 3 CLO1 C2

Client(cid, cname)

Contract(cn_id, client_id, contract_title, dateFrom, dateTo)

Write an assertion that will allow only one contract for a client from "19-Jun-2020" to "25-Jun-2020"

c) Given the following table of information 5 CLO1 C2

Employee (ID,cname,Age,Address,Salary, designation, dept)

Salary_pay(pid, ID, salary_of_month, pay_date)

Write a trigger that will automatically increase the salary (5%) of the employee if he is from "finance" department and receiving the salary of June 2020 and onwards.

OR

c) What are the different types of referential integrity constraints, and how can they be used to ensure the integrity of data in a relational database? 5 CLO1 C2

2 a) Explain the problem (with examples) if the data remain un-normalized. How could you resolve them? Explain with example 5 CLO2 C3

OR

a) What is an equivalence of sets of functional dependencies? 5 CLO2 C3

R=(A,B,C,D,E,F)

F1={A->BC, B->CDE, AE->F}

F2={A->BCF, B->DE, E->AB}

Check whether F1 and F2 are equivalent or not.

b) The following relation schema can be used to register information on the repayments on micro loans. 5 CLO2 C2

Repayment **(borrower_id, name, address, loanamount, requestdate, repayment_date, repayment_amount)**

A borrower is identified with an unique borrower_id, and has only one address. Borrowers can have multiple simultaneous loans, but they always have different request dates. The borrower can make multiple repayments on the same day, but not more than one repayment per loan per day.

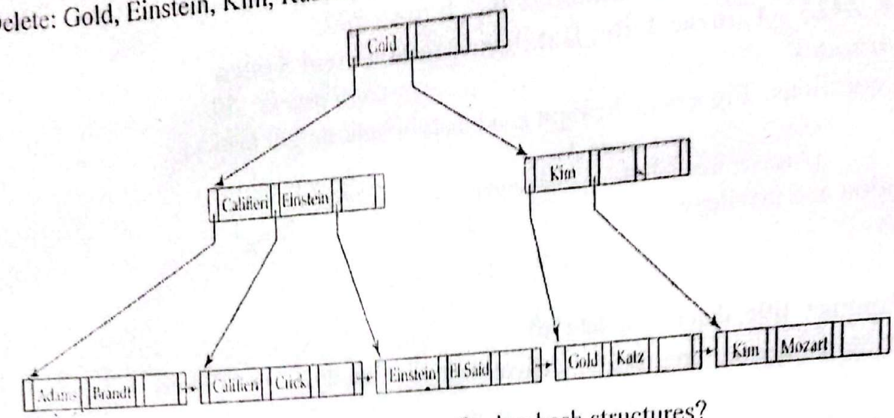
a) State a key (candidate key) for Repayment.
b) Make the normalization to BCNF. Show the steps.

Group B

[Answer the following questions]

3. a) What is the difference between closed hashing and open hashing in DBMS, and which one is generally more efficient? 2 CLO1 C2

b) Given the following B+-tree, do the following in it (Step by step): 5 CLO1
Insert: Marry, Hasi, Nicolous, Oli, Panalous, Quantas, Russel, Stain, Trucer, Zena
Delete: Gold, Einstein, Kim, Russel, Stain, Trucer, Zena



c) What are the advantages and disadvantages of using hash structures? 3 CLO1

OR

c) Construct a B+ tree for the following set of data: [5, 10, 15, 20, 25, 30, 35, 40, 45, 50]

4. Answer the following questions: 10 CLO1

1. List the ACID properties. Explain the usefulness of each
2. Justify the following statement: Concurrent execution of transactions is more important when data must be fetched from (slow) disk or when transactions are long, and is less important when data is in memory and transactions are very short.
3. Explain the distinction between the terms *serial schedule* and *serializable schedule*
4. Since every conflict-serializable schedule is view serializable, why do we emphasize conflict serializability rather than view serializability?

OR

Answer the following questions: 10 CLO1

1. Why is cascadelessness of schedules desirable? Are there any circumstances under which it would be desirable to allow noncascadeless schedules? Explain your answer
2. If deadlock is avoided by deadlock avoidance schemes, is starvation still possible? Explain your answer.
3. For each of the following protocols, describe aspects of practical applications that would lead you to suggest using the protocol, and aspects that would suggest not using the protocol:
   - Two-phase locking
   - The tree protocol
4. When a transaction is rolled back under timestamp ordering, it is assigned a new timestamp. Why can it not simply keep its old timestamp?

5. a) Explain the purpose of the checkpoint mechanism. How often should checkpoints be performed? How does the frequency of checkpoints affect 4 CLO1 C1
   - System performance when no failure occurs
   - The time it takes to recover from a system crash

b) When is it useful to have replication or fragmentation of data? Explain your answer.

c) What are the advantages and disadvantages of using multilevel indexing in databases? 2 CLO1 C2

d) What are the advantages of using a B+ tree index over a binary search tree index in a database? 2 CLO1 C2

Or 2 CLO3 C2

d) What are the advantages of using a B+ tree index over a B-tree index for a database?

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[Answer all the question; Figures in the right hand margin indicate full marks]

CO DL

1 (a). Consider the following database schema and answer the questions. 6 CO2 C3

Employee(E_id, C_id,E_Name,Email,Phone_no,Salary,Address,F_id)
Customer(C_ID, C_Name, Email,Phone_no,F_id)
Food_Item(F_ID, C_ID,F_Name, Price, Quantity)

1) Write the sql statement required to create these relations, you have to include all integrity constraints.
2) Define constraints on food item table as such quantity should be at least one item.
3) Define constraints on Employee such that a minimum 15000 BDT pay scale is followed by the organization.

1 (b). i) Define referential integrity constraints and domain constraints with example. 4 CO2 C3

ii) Describe the following:

a) Encryption and Decryption
b) Privileges..

2 (a). Consider the relation schema R = (A, B, C, D, and E) having following set *F* of functional dependencies: 8 CO2 C3

A→BC
CD→E
B→D
E→A

Requirements:

1. Suppose that we decompose the relation schema (A, B, C) and (A, D, E). How that this decomposition is a lossless decomposition if the above functional dependencies holds?
2. Give an example of a relation schema R and a set of dependencies such that R is in BCNF but is not in 4NF.
3. Compute the closure and list the candidate keys for R considering above functional dependencies.

OR

Briefly answer the following questions:

a) Give a set of FDs for the relation schema *R(A,B,C,D)* with primary key *AB* under which *R* is in 1NF but not in 2NF.
b) Give a set of FDs for the relation schema *R (A,B,C,D)* with primary key *AB* under which *R* is in 2NF but not in 3NF.
c) Consider the relation schema *R(A,B,C)*, which has the FD *B → C*. If *A* is a candidate key for *R*, is it possible for *R* to be in BCNF? If so, under what conditions? If not, explain why not.

2 (b). What do you understand by functional dependency? Write down and explain the types of functional dependency. 2 CO

OR

What do you mean by Normalization? Why normalization is important?

GROUP-B

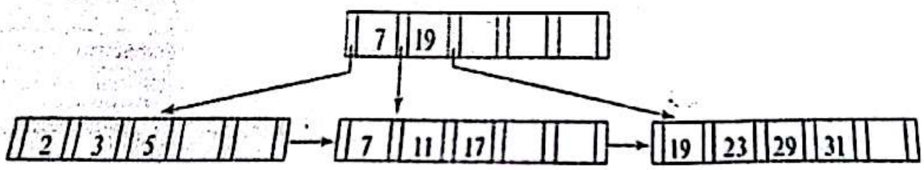
3 (a). 1) Explain the distinction between closed and open hashing. Discuss the relative merits of each technique in database applications. 6 CO2 C2

2) What are the causes of bucket overflow in a hash file organization? What can be done to reduce the occurrence of bucket overflows?

3) Why is a hash structure not the best choice for a search key on which range queries are likely?

3(b). Consider following B+ tree and perform following operations step-by-step in it 4 CO2 C3

1. Insert 8 2. Insert 4 3. Insert 13 4. Insert 20 5. Insert 22
6. Insert 35 7. Delete 3 8. Delete 20 9. Delete 19 10. Delete 29



4. *Answer the following questions:* 10 CO1 C2

1. *Describe how a typical lock manager is implemented. Why must lock and unlock be atomic operations?*
2. Contrast the timestamps assigned to restarted transactions when timestamps are used for deadlock prevention versus when timestamps are used for concurrency control.
3. Show that, if two schedules are conflict equivalent, then they are view equivalent.
4. Give an example of a serializable schedule that is not strict.

OR

Answer the following questions:

1. Show that the two-phase locking protocol ensures conflict serializability and that transactions can be serialized according to their lock points.
2. What benefits does rigorous two-phase locking protocol provide? How does it compare with other forms of two-phase locking?
3. Show by example that there are schedules possible under the tree protocols that are not possible under the two-phase locking protocol, and vice versa.
4. Why should transactions need to assure ACID properties?

5(a). What is B+ tree? Compare between primary and secondary indices. 3 CO1 C2
5(b). When does multilevel indexing preferable? Justify your answer with example. 3 CO1 C2
5 (c). Construct B+ tree for the following set of key values. 4 CO1 C2
1,6,8,12,15,19,14,18,32,40,51,46,60,55,62
Assume that the tree is initially empty and Node Size: 4

** THE END**

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B. Sc. in CSE Final Examination, Autumn2022

Course Code: CSE 2423 **Course Title: Database Management System**

Total marks: 30

Time: 2 hours30 minutes

[Answer all the questions; in some questions, there are options;solve the one which you have been instructed to solve;Precisely follow the guideline for preparing and submitting the answer script;Figures in the right-hand margin indicate full marks]

Course Outcomes (COs) of the Questions	
01	**Understand**Relational Databases, Database design, Data Storage and Querying, Transaction Management.
02	**Apply** Relational Algebra, SQL, Query Optimization techniques, Data Integrity, Security, normalization techniques, Indexing Techniques, and ACID Properties.
03	**Create**an enterprise data model that reflects the organization's fundamental business rules.

Question	Marks	CO	DL
Consider the following relational schema and briefly answer the questions that follow: Emp(*eid:* integer, *ename:* string, *age:* integer, *salary:* real) Works(*eid:* integer, *did:* integer, *pct time:* integer) Dept(*did:* integer, *budget:* real, *managerid:* integer) 1. Write the SQL statements required to create these relations, including all integrity constraints such as primary and foreign keys. 2. Define a domain constraint on Emp that will ensure that every employee makes at least \$10,000. 3. Define an assertion on Dept that ensure that all managers have *age* >30.	6	CO2	Ap
Define referential integrity. Explain the tests that must be made to preserve referential integrity for ***delete*** operation.	4	CO1	Un
Suppose you are given a relation *R* with four attributes *ABCD*. Foreach of the following sets of FDs, assuming those are the only dependencies that holdfor *R*, do the following: a) Identify the candidate key(s) for *R*. b) Identify the types of functional dependencies (FD) that exist. Why do you think those are FDs? c) Identify the bestnormal form that *R* satisfies (1NF, 2NF, 3NF, or BCNF). d) If *R* is not in BCNF, decompose it into a set of BCNF relations that preserve the dependencies. 1. $C \rightarrow D, C \rightarrow A, B \rightarrow C$ 2. $B \rightarrow C, D \rightarrow A$ 3. $ABC \rightarrow D, D \rightarrow A$ 4. $A \rightarrow B, BC \rightarrow D, A \rightarrow C$ 5. $AB \rightarrow C, AB \rightarrow D, C \rightarrow A, D \rightarrow B$	10	CO2	Ap

OR

2. When a relational schema will be in 1NF, 2NF, and 3NF. Illustrate with an example how you can convert the following Schema ***Sales Record*** into 3NF. **10** CO2

CustName	**Item**	**ShippingAdd**	**Newsletter**	**Supplier**	**SupplierPhone**	**Price**
Hasan	Galaxy	3/1 Ctg	Gal-News	Samsung	22-8805568	25000
Kamal	iPhone	2-A Dhk	I-News	Apple	66-5668412	65000
Sojib	Galaxy, iPhone	255 Raj	Gal-News, I-News	Wholesale	Toll-Free	90000
Hasan	iPhone	1/6A Syl	I-News	Apple	66-5668412	65000

GROUP-B

3. (a) Construct a B+-tree for the following set of key values 5 CO2
6, 3, 2, 7, 8, 12,18, 17, 19, 23, 29, 31, 35, 38, 40, 50, 45, 56, 58, 62, 70, 73, 71, 75, 77, 85, 81, 82
Assume that the tree is initially empty and **Node** Size: Four

3. (b) Explain how you will assess the quality of an index using index evaluation metrics. 5 CO1
Compare between:

1. ClusteringandNon-Clustering indexes.
2. Dense and Sparse indexes.

4. (a) What is a transaction? Explain its ACID properties with examples. 4 CO1

4. (b) Explain how lock-based **protocol** supports concurrency control. Illustrate how deadlock and starvation occur in the lock-based protocol. 6 CO1

OR

4.

a) What is the phantom problem? Can it occur in a database where the set of database objects are fixed and only the values of objects can be changed? 6 CO1

b) Consider a database with objects X and Y and assume that there are two transactions $T1$ and $T2$. Transaction $T1$ reads objects X and Y and then writes object X. Transaction $T2$ reads objects X and Y and then writes objects X and Y. 4 CO1

1. Give an example schedule with actions of transactions $T1$ and $T2$ on objects X and Y that results in a write-read conflict.
2. Give an example schedule with actions of transactions $T1$ and $T2$ on objects and Y that result in a read-write conflict.
3. Give an example schedule with actions of transactions $T1$ and $T2$ on objects X and Y that results in a write-write conflict.
4. For each of the three schedules, show that Strict 2PL disallows the schedule.

5. (a) Draw the state diagram of a transaction and describe its states with an example. Explain how the shadow copytechnique supports durability. 5 CO1

5. (b) Write down the differences between log-based recovery and checkpoint-based recovery with appropriate examples. 5 CO1

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Course Code: CSE-2423
Time: 2 hours 30 minutes

Course Title: Database Management System
Full Marks: 50

(i) The figures in the right-hand margin indicate full marks
(ii) Course Outcomes and Bloom's Levels are mentioned in additional Columns

Course Outcomes (COs) of the Questions	
CO1	Understand Relational Databases, Database design, Data Storage and Querying, Transaction Management.
CO2	Apply Relational Algebra, SQL, Query Optimization techniques, Data Integrity, Security, normalization techniques, Indexing Techniques, ACID Properties.
CO3	Create an enterprise data model that reflects the organization's fundamental business rules.

Bloom's Levels of the Questions						
Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Part A

[Answer the questions from the followings]

1. a) Consider the following relational database: CO1 U

Patient (pid, pname, Address, mobile, DOB, gender)
visit (pid, did, visit_date)
Doctor (did, dname, speciality)
drug (drid, d_name, d_type, manuf_year, unit_price)
prescribe (pid, did, drid, pdate, quantity)

Give an SQL DDL definition of this database. Identify referential-integrity constraints that should hold, and include them in the DDL definition.
Ensure the following constraints:

Pname, dname, and d_name (not null), Mobile (unique), gender ('M', or 'F'), quantity and unit_price (not negative).

OR

1. a) What is referential integrity? Explain the tests that must be made to preserve referential integrity for update operations. CO1 U

1. b) Write an assertion using the schema from 1(a) that will allow only 'female' patient to visit the 'Gynecology' specialist. CO2 Ap

2. a) Consider a single example from your own, identify different anomalies and functional dependencies in it and apply different normalization techniques to resolve the anomalies and functional dependencies. CO2 Ap

OR

2. a) When a relational schema will be in 1NF, 2NF, and 3NF. Illustrate with an example how you can convert a Schema ***Student*** with multi-valued attribute "email" into 1NF. — CO2 Ap 5

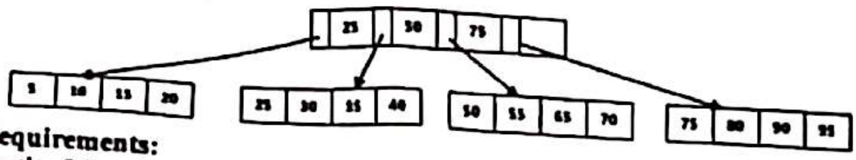
2. b) 1. How does different authorization techniques could be deployed to the user and revoked? Explain with example. — CO1 U 3

2. List out different encryption techniques obtained so far? Which one is preferable, why? — 2

Part B

[Answer the questions from the followings]

3. a) Given the following B+ tree — CO2 Ap 6



Requirements:

Do the following (step by step)

Insert: 13,12,17, 60, 45

Delete: 35, 60, 75, 95,13

OR

a) Define domain constraint. Create a domain constraint *min-age* for an employee table in a company database that will not allow entering employees less than 18 years old. — CO2 C 6

b) How can Indexes help performance? Consider employees relation, if we want to retrieve all employees, whose salary is in a given range, will it be best alternative to sort the employee records by employee id. Justify your answer. — CO1 U 4

a) Describe multilevel indexing with a suitable example and necessary figure. — CO1 U 4

OR

a) How could you resolve the problem of skew? Explain with example. — CO1 U 4

b) How a typical lock manager works using compatibility matrix? Why must lock and unlock be atomic operations? What is starvation and how should a lock manager handle it? Explain each question with example. — CO1 U 6

a) What is a transaction? Explain its ACID properties with examples. — CO1 U 4

b) Draw the state diagram of a transaction and explain. Explain how shadow copy technique works. — CO1 An 6

Create assertion pat-test {

check (not exists ((select * from

Patient where gender = "F" and

(select * from Doctor where speciality

= "gynecology"))